Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years building enterprise-grade data pipelines and data warehouses at scale. Owned end-to-end ETL pipelines transforming multi-source financial and operational datasets into 180,000+ domain records powering KPI calculations, financial reconciliation, and reporting for 100K+ monthly active users -- with dimensional modeling, Medallion-style layered architecture, rigorous data quality frameworks, and zero compliance violations across 7 years. Optimized 200+ SQL queries (30% gain) across Oracle, SQL Server, and PostgreSQL using CTEs, window functions, and execution plan analysis. Strong Azure ecosystem experience (Databricks, Data Factory, ADLS) and proficiency across Snowflake, Teradata, Python, Scala, PySpark, and Apache Spark for large-scale data pipeline engineering. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Azure Databricks + ADF + ADLS | Snowflake + Teradata + Oracle SQL | Dimensional Modeling + Medallion Architecture | End-to-End ETL Pipeline Design | Financial Reconciliation + KPI Pipelines | Data Quality Frameworks + Root Cause Analysis | Python + Scala + PySpark + Apache Spark | Multi-Source Data Integration + Validation

WORK EXPERIENCE

Entrata (PropTech / Real Estate SaaS)

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned end-to-end ETL data pipeline for 7 years** -- transformed HUD federal source data into ~180,000 domain records across a dimensional data warehouse powering financial KPI calculations, eligibility, rent, and tax-credit reporting for 100K+ housing units. Medallion-style layering: raw ingestion, validated intermediate tables, curated domain records with data quality frameworks (pre-import baselines, reconciliation counts, rollback preparation). Zero compliance violations across all annual release cycles.
- **Optimized 200+ SQL queries achieving 30% performance gain** -- execution plan analysis, indexing, partitioning, CTEs, window functions, and stored procedure refactoring across Oracle, PostgreSQL, and SQL Server for financial reconciliation and operational KPI workloads. Root cause analysis on data anomalies and incorrect financial calculations across large, complex multi-tenant datasets.
- **Designed data warehouse schemas and data marts** supporting KPI dashboards and financial reporting -- dimensional modeling, normalization and denormalization, star schema structures aligned to reporting requirements, business modeling, and scalability across multi-tenant environments serving 100K+ MAU.
- **Led cloud data migration to AWS S3** with encrypted ETL pipelines (100% PHI compliance, Libsodium) -- inventoried all data assets, mapped legacy records to target schemas, validated migrated content, confirmed downstream financial workflows remained accurate. Zero security incidents.
- **Built Python and PySpark automation pipelines** eliminating 80% of manual operational workload; managed delivery across 120+ Agile sprints using Jira backlogs, Git version control, and iterative development practices.

George Mason University

Aug 2024 - Aug 2025

Research Associate · College of Engineering and Computing

- Built multi-source data pipelines (Python, R) over WHO and UK Road Safety datasets (190+ countries) -- ingestion, transformation, validation, root cause analysis on data quality issues via Isolation Forest anomaly detection. R2 = 0.977 regression accuracy.
- Designed Tableau, Power BI, and ggplot2 KPI dashboards; fine-tuned classification models (HuggingFace, PyTorch) with offline evaluation harnesses.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built automated data extraction and validation pipelines integrating FAA regulatory APIs; engineered data quality guardrails enforcing integrity checks before processing. Python + LangGraph orchestration.

Entrata India, Pune

Jul 2017 - Sep 2018

Associate Software Engineer

- Built income-limit import ETL pipelines processing 1.6M HUD/Tax Credit records; led cloud migration to AWS S3; redesigned data storage database architecture migrating hundreds of thousands of records to a modern schema with 90% legacy space resolution.

PROJECTS

Leasel: Enterprise Data Platform for Compliance Operations

Python · LangGraph · FastAPI · SQLite · ETL

Built enterprise data platform consolidating compliance documents, regulatory rules, system settings, historical tickets, and release notes into a unified data mart. ETL ingestion pipeline with OCR fallback, section-aware chunking, TF-IDF retrieval with citation grounding. Applied data quality frameworks with validation gates -- every answer cited or flagged unsupported. 8-agent orchestration for root-cause analysis across large, complex compliance datasets. Results: 93.6% precision, 87.9% F1 across 1,240 lease PDFs (5 US states).

AgentProbe: Evaluation Pipeline with KPI Dashboards

Python · FastAPI · CI/CD · Golden Datasets

End-to-end evaluation pipeline with golden datasets, regression harnesses, and 5-metric KPI engine (correctness, groundedness, latency, token efficiency, hallucination rate). CI-integrated data quality gate with root cause analysis on regressions. github.com/venkatalakshmiparimala/AgentProbe

CrashClarity: Large-Scale Road Accident Analysis (104K Records)

Python · R · scikit-learn · Tableau

End-to-end data pipeline on 104K+ UK accident records -- multi-source ingestion, feature engineering, Random Forest + Logistic Regression (85%+ accuracy), actionable KPI dashboards deployed on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87 I Coursework: Data Warehousing, ETL Pipeline Design, Statistical Modeling, Machine Learning, Data Visualization

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Languages: Python, SQL, PL/SQL, Java, Scala, JavaScript, TypeScript, R, Bash

Data Engineering: Azure Databricks, Azure Data Factory (ADF), Azure Data Lake Storage (ADLS), Snowflake, Teradata, Apache Spark, PySpark, AWS Glue, ETL, Data Warehouse Design, Data Mart Development, Dimensional Modeling, Medallion Architecture, Star Schema, Financial Reconciliation, KPI Pipeline Design, Data Lineage, Data Governance

Databases: PostgreSQL, Oracle, Microsoft SQL Server (T-SQL), MySQL, Snowflake, Teradata, NoSQL, Redis

Cloud & Infra: Azure (App Service, SQL, ADF, ADLS, Databricks), AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, RabbitMQ, CI/CD (GitHub Actions), Jira, Agile/Scrum, Git

Data Quality: Data Quality Frameworks, Root Cause Analysis, Reconciliation Validation, Performance Tuning (CTE, Window Functions, Execution Plan Analysis, Query Optimization, Indexing, Partitioning)

Analytics & ML: Tableau, Power BI, Pandas, NumPy, Scikit-learn, XGBoost, Isolation Forest, Anomaly Detection, Statistical Modeling, HuggingFace, PyTorch